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MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, HERMETICALLY SEALED, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for hermetically sealed, electrical-resistance, temperature indicators.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric "AN" TYPE
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connector, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

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MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

* 3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2.2 Nonmagnetic metals. Nonmagnetic metals shall be used for all parts except where magnetic metals are essential.

3.3.2.3 Nonferrous metals. Nonferrous metals shall be used for all parts of the indicator except where ferrous metals are essential.

3.3.3 Corrosive fumes. Materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

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3.3.4 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the perforance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed to indicate temperature when used in conjunction with an electrical-resistance circuit. It shall be so constructed as to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Pivots, bearings, and gears. Pivots, bearings, and gears shall not bind nor shake and shall be as frictionless as practicable.

3.4.2 Case. The dimensions of the case shall be as specified in the detail specification. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity.

* 3.4.2.1 Hermetic sealing. The case shall be constructed of a material suitable for hermetic sealing. The material shall be uniform in texture and shall have a smooth external surface. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.2.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 90 percent nitrogen and 10 percent helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

3.4.2.3 Case finish. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in flat black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental conditions specified herein.

* 3.4.3 Dial. The dial shall be as specified in the detail specification and shall consist of a flat disc of metal or plastic material. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn under vibration. For each indicator part number, there shall be only one part number for the dial.

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3.4.3.1 Dial marking. The dial shall be marked as specified in the detail specification. The form of the numerals and the letters shall conform to MS33558.

* 3.4.4 Pointer. The pointer design shall be as specified in the detail specification. The pointer shall be as light as practicable and shall be sufficiently rigid to prevent flexing under vibration. The length shall be such that the portion of the pointer that is visible will be a minimum of 1/2 inch and the pointer tip will extend into the scale a distance equal to one-third to two-thirds the length of the shortest graduation. The pointer on nonlighted indicators shall be channeled to provide adequate strength. For lighted indicators, pointers shall be of suitable design for integral lighting.

3.4.4.1 Pointer overswing. Pointer stops shall be provided and shall be so located as to permit the pointer to overswing the amount specified in the detail specification.

3.4.4.2 Parallax. To minimize parallax, the clearance between the pointer and the plane of the dial face shall not exceed 0.10 inch.

3.4.4.3 Pointer marking. The pointer shall be marked as specified in the detail specification.

3.4.5 Cover glass. The cover glass shall be suitable for hermetic sealing and shall be flat and free from discolorations, scratches, and striae which would interfere with normal reading of the indicator.

* 3.4.5.1 Dial to cover glass distance. The distance between the inside of the coverglass and the dial shall not exceed 0.140 inch for nonlighted indicators nor 0.188 inch for integrally lighted indicators.

3.4.6 Mask. A thin metal mask, as shown in the detail specification, shall be provided directly behind the cover glass to cover all moving parts of the mechanism, except the pointer, and to provide for a uniform visible length of pointer regardless of the pointer indication. The mask shall be finished in durable dull black.

* 3.4.7 Reliability. When reliability is required, it shall be in accordance with MIL-STD-781 and as specified in the detail specification.

3.5 Performance. The indicator shall be capable of operating as specified herein under the following conditions:

a. Temperature - Temperature ranging from -65° to +71°C

b. Sealing - Immersion in water with a pressure of 2-1/2 inches Hg applied for a period of 1 minute

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c. Open circuit and short circuit - Open circuiting and short circuiting the resistance bulb circuit 5 times each

* d. Vibration - A double amplitude of 0.10 inch from 5 to 20 Hz and vibration of 2g from 20 to 500 Hz

* e. Salt fog - Exposure to simulated salt sea atmosphere for a period of 48 hours.

3.5.1 Scale error. Scale error and tolerances shall be as specified in the detail specification.

* 3.5.2 Friction error. Friction error for increasing and decreasing scale readings shall not exceed the tolerances specified in the detail specification.

* 3.5.3 Position error. Position error shall not exceed the tolerances specified in the detail specification when the indicator is rotated clockwise (cw) and counterclockwise (ccw) or when tilted forward and backward 45° from the normal position.

3.5.4 Insulation resistance. Insulation resistance shall be as specified in the detail specification when a potential of 150V dc is applied between any electrical pin contact and the shell, shielding, or any metal part of the case.

* 3.5.5 Magnetic effect. When the indicator is revolved about a short bar magnetic compass with a field intensity of 0.17 to 0.19 oersted, deflection of the compass shall not exceed the value specified in the detail specification.

* 3.5.6 Resistance bulb current. The current consumed in the resistance bulb circuit shall not exceed the value specified in the detail specification.

* 3.5.7 Thermal shock and fogging. The indicator shall withstand eight cycles of submersion in tap water at 85° ±2°C and then in water at 5° ±2°C as specified in 4.6.20 without evidence of moisture penetration, condensation, fogging, or damage to the external finish or enclosure.

* 3.6 Part numbering of interchangeable parts. All parts having the same manufactuer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern the manufactuer's part numbers and changes thereto.

* 3.7 Lighted indicators. Integrally lighted indicators shall conform to MIL-L-25467 or MIL-L-27160, as applicable.

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3.8 Balancing. The moving element of the indicator shall be balanced by an approved method (such as threaded lock or split nuts or a wire helix moving along a cross arm and counterweight arm) and shall have an easily accessible means for rebalancing. Solder, shellac, or similar means of holding balance weights shall not be used.

3.9 Bearings and pivots. Moving elements shall turn on hardened, accurately ground and highly polished pivots bearing on highly polished first-quality sapphire bearings or other materials, as approved by the qualifying activity, as alternates to precious gems.

* 3.10 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.11 Wiring. Wherever practicable, internal wiring shall consist of tinned flexible stranded wire.

3.12 Electrical receptacle. The electrical receptacle shall be in accordance with MIL-C-5015 or MIL-C-26482 and as specified in the detail specification. Electrical receptacles of the hermetically sealed type not covered by MIL-C-5015 but having the dimensional characteristics of those receptacles may be used provided prior approval from the procuring activity is obtained.

* 3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

* 3.14 Marking. Markings shall be finished in white, color No. 37875 of FED-STD-595, and shall be so durable as to withstand usage encountered in service.

3.14.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.15.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.15.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt can not be appreciably tightened further without damage or injury to the screw, bolt, or threads.

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3.15.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.15.4 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.15.5 Cleaning. During various stages of assembly, the indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips and other foreign material, burrs, sharp edges, and resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the indicator shall be tapped lightly before a test reading is taken (see 6.3.1).

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

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4.4 Qualification testing

4.4.1 Test samples. The qualification test samples shall consist of three indicators. The samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.4.2 Qualification tests. The qualification test shall consist of all the tests specified herein.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

* 4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Scale error at room temperature

c. Position error

d. Friction error

e. Sealing

f. Insulation resistance

g. Individual tests of applicable lighting specification.

4.5.2 Sampling tests

* 4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Reduced voltage operation

c. Scale error at low temperature

d. Scale error at high temperature

e. Magnetic effect

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f. Resistance bulb current

g. Open circuit and short circuit

h. Vibration

i. Sampling plan A tests of applicable lighting specification.

* 4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 indicators shall be selected at random from the first 15 produced on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Response time

c. Damping factor

d. Power consumption

e. Bulb lead resistance effect

f. Extreme low temperature exposure

g. High temperature exposure

h. Thermal shock and fogging

i. Salt fog

j. Internal examination

k. Sampling plan B test of applicable lighting specification.

4.5.2.3 Rejection and retest. When one indicator selected from a production run fails to meet the specification, no indicators still on hand or later produced shall be accepted until the extent and cause of failure are determined and the defects have been corrected. After corrections have been made, all necessary tests shall be repeated.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, case diameter, and dimensions.

4.6.2 Scale error at room temperature. The resistance specified in the detail specification shall be connected to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors shall be determined at the test points specified in the detail specification for the indicator under test. The scale errors shall not exceed the tolerances specified therein.

4.6.3 Position error. The reading of the indicator shall be observed in the normal operating position (plane of dial vertical) with no voltage applied. The pointer shall be off scale at the low-temperature end of the scale. The indicator shall then be rotated 90° cw and 90° ccw, maintaining the plane of the dial vertical. The indicator shall then be tipped forward and backward 45° from the normal operating position. The change in pointer indication produced by these positions shall be noted while the indicator is being lightly tapped. The pointer shall remain slightly off scale at the low-temperature end of the scale. This test shall be repeated with test voltage and resistance applied to produce an indication, and may be performed at any indication at the discretion of the qualifying activity. The maximum change in reading with voltage applied, as produced by the test position, shall not exceed the tolerance specified in the detail specification. When power to the indicator is removed, the indicator pointer shall return to the on-scale position without tapping.

4.6.4 Friction error. This test may be conducted in conjunction with the scale error test. Differences in reading shall be noted after a slow change in increasing indication before and after the indicator is lightly tapped. The procedure shall be repeated for decreasing indication. This test may be performed at any point on the scale at the discretion of the qualifying activity. The maximum observed difference in reading shall not exceed the tolerance specified in the detail specification. The movement of the pointer shall be smooth when the bulb resistance is varied uniformly.

4.6.5 Sealing. The indicator shall be immersed in a suitable liquid, such as water. The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch Hg and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is longer. The absolute pressure shall then be increased to 2-1/2 inches Hg. Any bubbles coming from within the indicator case shall be considered as leakage and shall be cause for rejection; however, bubbles that are the result of entrapped air on the various parts of the case shall not be considered as leakage. A helium leak detector or other means of testing that is equal or superior in sensitivity to

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this test method may be used upon approval by the qualifying activity. Where a leak detector is used for conducting this test, the leak rate shall not exceed 0.1 micron-cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.6 Insulation resistance. A potential of 150V dc shall be applied between any electrical receptacle pin contact and the shell, shielding, or any metal part of the indicator case. The insulation resistance shall be as specified in the detail specification.

4.6.7 Reduced voltage operation. The indicator shall be subjected to this test with the applied voltage specified in the detail specification. The indications at the reduced voltages shall not differ from the indications at design voltage by more than the tolerances specified in the detail specification.

4.6.8 Scale error at low temperature. The indicator shall be placed in a temperature chamber and subjected to -54° ±2°C for a period of 4 hours. At the end of this period and while still at the low temperature, resistance and voltage as specified in 4.6.2 shall be applied to the indicator. The errors at low temperature shall not exceed the tolerances specified in the detail specification.

4.6.9 Scale error at high temperature. The indicator shall be placed in a temperature chamber and subjected to 71 ±2°C for a period of 4 hours. At the end of this period and while still at the high temperature, resistance and voltage as specified in 4.6.2 shall be applied to the indicator. The errors at high temperature shall not exceed the tolerances specified in the detail specification.

4.6.10 Magnetic effect. The indicator, first not operating and then operating on rated voltage, shall be revolved in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed the tolerances specified in the detail specification.

4.6.11 Resistance bulb current. The test voltage and bulb resistance specified in the detail specification shall be applied to the indicator and the current in the resistance bulb circuit shall be measured. The current consumed in the resistance bulb circuit shall be as small as practicable and shall not exceed the tolerances specified in the detail specification.

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4.6.12 Open circuit and short circuit. The test voltage specified in the detail specification shall be applied to the indicator and the resistance bulb circuit shall be open circuited five times and short circuited five times causing the pointer to go off scale five times in each direction. The duration of each open circuit shall be 30 seconds. Following this test, the indicator shall be subjected to the scale error at room temperature test. The scale errors at room temperature after open and short circuit shall not exceed the tolerances specified in the detail specification.

* 4.6.13 Vibration. The vibration test shall be conducted in accordance with MIL-STD-810, method 514, equipment category (a), curve B.

4.6.13.1 Pointer oscillation. The maximum total amplitude of pointer oscillation throughout the frequency range shall be noted and shall not exceed the amount specified in the detail specification. This test may be conducted at any indication at the discretion of the qualifying activity.

4.6.13.2 Pointer variation. During vibration, the maximum variation of the pointer from its original position before vibration, throughout the frequency range, shall be noted and shall not exceed the amount specified in the detail specification. This test may be conducted at any indication at the discretion of the qualifying activity.

* 4.6.14 Response time. Voltage and resistance sufficient to produce momentary two-thirds full-scale deflection shall be applied to the indicator. The length of time, in seconds, required for the pointer to come to rest shall be noted. This test shall be repeated several times and the average value taken as the response time of the indicator. The pointer shall be considered at apparent rest when it has reached within ±1 percent of the scale length of the actual rest point. The response time shall not exceed the tolerance specified in the detail specification.

4.6.15 Damping factor. The test voltage shall be applied to the indicator and two suitable adjustable resistances, R_1 and R_2, connected in series across the resistance bulb connections. A short circuiting switch shall be connected across R_2. R_1 shall be adjusted so that the pointer reads at the lowest scale graduation when R_2 is short circuited. R_2 shall then be adjusted so that the pointer momentarily deflects to the top scale graduation when the short circuit is removed. The steady state and maximum momentary deflection values shall be recorded. The damping factor shall be taken as the ratio of the steady-state deflection in angular degrees to the difference between the maximum momentary deflection and steady-state deflection in angular degrees. The damping factor shall not exceed the value specified in the detail specification.

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* 4.6.16 Power consumption. The power consumption of the indicator shall be measured with the test voltage and sufficient resistance applied to the indicator to produce a full-scale deflection and recorded in terms of watts. The power consumption of the indicator shall not exceed the value specified in the detail specification. The power consumption shall not include power for integrally lighted indicators.

4.6.17 Bulb lead resistance effect. The indicator shall be connected as specified in the detail specification and subjected to the scale error at low temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in the detail specification.

4.6.18 Extreme low temperature exposure. The indicator shall be subjected to a temperature of -65° ±2°C for a period of 48 hours. The temperature shall then be raised to -54° ±2°C for a period of 24 hours. During the last 4 hours of the 24-hour period at -54°C, the pressure shall be reduced to 0.82 inch Hg absolute, or less. Upon completion of the 48-hour period, and again at the completion of the 24-hour period, while at the specified temperature and pressure, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. Upon completion of the 24-hour period, while at -54°C but at atmospheric pressure, the indicator shall be subjected to the test specified in 4.6.8. The scale errors at the test points shall not exceed the tolerances specified in the detail specification. The indicator shall then be returned to room temperature for a minimum period of 4 hours and then subjected to the test specified in 4.6.2. The scale errors at the test points shall not exceed the tolerances specified in the detail specification.

4.6.19 High temperature exposure. The indicator shall be placed within a chamber and the internal temperature of the chamber raised to 71° ±2°C with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat will not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at 71° ±2°C for a period of 24 hours. While still at this temperature, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The indicator shall then be returned to room temperature for a minimum period of 4 hours and then subjected to the test specified in 4.6.2. The scale errors at the test points shall not exceed the tolerances specified in the detail specification.

* 4.6.20 Thermal shock and fogging. The indicator shall be subjected to eight cycles of submersion in tap water at 85° ±2°C and 5° ±2°C. There shall be no evidence of moisture penetration, condensation, or damage to the external finish or enclosure. Each cycle shall consist of immersing the indicator in tap water at 85° ±2°C for a period of 30 minutes and within 5 seconds immersing it in the

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5° ±2°C tap water for a period of 30 minutes. During the fourth and eighth cycles, at the end of the 30-minute soak in 85°C tap water, the indicator shall be removed from the hot bath and within 4 seconds the face of the instrument inserted in the 5°C tap water to a depth of 1/4 ±1/8 inch. At the end of 1 minute, the indicator shall be withdrawn from the 5°C bath and the cover glass immediately wiped with a cloth or paper product and observed for internal fogging. If no fogging is observed, the sequence of the test shall be continued to completion.

* 4.6.21 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810, method 509, for a period of 48 hours. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which would affect subsequent operation. The indicator shall then be resubjected to and shall meet the insulation leakage and sealing tests. No indicator leakage shall result from this test. The external surface of the indicator may be washed with distilled water and air dried prior to conducting the insulation leakage test.

4.6.22 Internal examination. The indicator shall be examined internally for evidence of corrosion or deterioration resulting from the tests specified herein. It shall also be critically examined to determine full compliance with regard to weight, physical dimensions, dial composition, and similar detail requirements. There shall be no evidence of corrosion or deterioration.

4.6.23 Lighting. Lighting tests shall be conducted in accordance with MIL-L-25467 or MIL-L-27160, as applicable, in the order listed therein.

4.7 Inspection of preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, and packing. Indicators shall be preserved in accordance with MIL-P-116, method III, and packaged and packed in accordance with MIL-STD-794.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as specified in the detail specification.

* 5.2.1 Reinspection date markings. The reinspection date markings shall be as specified by the procuring activity (see 6.2).

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6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in aircraft for measuring various functions of temperature.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. Levels of packaging and packing required and reinspection date markings.

6.3 Definitions

6.3.1 Tapping. The phrase "tapped lightly" means the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - (AV)
Navy - (AS)
Air Force - 11

Review activities:
Army -
Navy -
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 6685-0213

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